Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Limoli, Charles Ph.D.		
Project Title:	High LET Radiation And Neurogenesis: Implications And Mechanisms Underlying Cognitive Impairment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
Start Date:	09/01/2003	**End Date:**	05/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: Original end date of 8/31/2007 was changed.		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1632		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	For those engaged in space travel, the radiation environment poses a range of potential health risks. For instance, it is possible that exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can induce a range of CNS injuries, including cognitive impairment. While the pathogenesis of radiation-induced cognitive impairment is not clear, we have shown that neuronal precursor cells in the hippocampus may be involved. Using low doses of X-rays (1-5 Gy), we have demonstrated, in vitro and in vivo, that neural stem cells of the hippocampus are exquisitely sensitive to irradiation. In addition, these cells exhibit oxidative stress and undergo apoptosis after irradiation. Low dose x-irradiation of these cells in vitro also leads to a persistent oxidative stress that coincides temporally with a dose-dependent reduction in neurogenesis observed in vivo. Consequently, we hypothesize that oxidative stress plays a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo. Further, we suggest that oxidative stress constitutes a biochemical mechanism that can influence the fate of stem cell differentiation in vitro and hippocampal neurogenesis in vivo. It is our overall objective to determine how oxidative stress affects the high LET radiation response of precursor cells in the subgranular zone (SGZ) of the hippocampus and to identify compounds/strategies that will enable us to ameliorate the adverse effects of high LET irradiation on hippocampal neurogenesis. We have defined 4 specific aims to address our objective. Aim 1: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to acute (days) oxidative stress and altered neurogenesis after exposure to high-energy protons. Aim 2: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to chronic (months) oxidative stress and altered differentiation/neurogenesis following exposure to high-energy iron ions. Aim 3: Determine if antioxidant treatment with SOD/catalase mimetic agents reduce high-LET induced radiation injury to neural stem cells in vitro and ameliorate the radiation-induced inhibition of neurogenesis in vivo. Aim 4: Quantify the effects of high-energy iron ions on cognitive function in wild type mice and determine if antioxidant treatment ameliorates those effects. Determine if or how oxidative stress impacts the radioresponse of neural precursor cell function. Elucidating oxidative stress as a biochemical mechanism regulating the behavior of neural precursor cells will provide significant insight into how irradiation might lead to impaired cognition during spaceflight. These studies will also provide information relevant to the development of antioxidant strategies to ameliorate the adverse effects of irradiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will help define the risks associated with radiation exposure in space. By understanding these risks NASA can improve its strategies for promoting the health of astronauts.
Task Progress:	Significant progress is summarized below: 1) Proton irradiation of neural precursor cells led to a rapid and dose-dependent increase in reactive oxygen species (ROS). 2) At 6h and 24h after irradiation, the relative increase in ROS was 10% and 3% per Gy respectively. 3) Protons led to a more rapid induction of ROS than X-rays, with an RBE of 1.06 and 1.23 after 1 and 5 Gy respectively. 4) Factors and/or cellular subpopulations responsible for inducing ROS were depleted by 2-days following proton irradiation. 5) Following proton irradiation, acute changes (6-24h) in mitochondrial function and content were not found, while significant decreases in mitochondrial content were observed 2 days later. 6) Exposure to iron ions (1-3 Gy) led to significant and dose-dependent reductions in neurogenesis, as measured by the loss of proliferating cells (BrdU and Ki67 positive) and their progeny immature neurons (doublecortin positive) 2-3 months after irradiation.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Rola, R.; Otsuka, S.; Obenaus, A.: Nelson, G.A.; Limoli, C.L.; VandenBerg, S.R.; Fike, J.R. "Indicators of hippocampal neurogenesis are altered by 56Fe-particle irradiation in a dose-dependent manner." Radiation Research 162(4):442-446, Oct 2004. , Oct-2004